HOLY ANGEL UNIVERSITY

SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS

A Capstone Project
Presented to the Faculty of the
School of Computing
Holy Angel University

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology with Area of Specialization in Web Development

November Student
Oscar Student
Hotel Student
India Student

April 2027

Final manuscript — revised after the final oral defense (see Summary of Revisions).
Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are placeholders.

SCHOOL OF COMPUTING

APPROVAL SHEET

This capstone entitled **"SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development has been examined and is recommended for acceptance and oral examination.

Prof. Charlie
Adviser

ORAL EXAMINATION

Approved by the committee of oral examiners on
April 12, 2027

Prof. Alpha
Panel Chair

AD Echo
Panel Member

[Name of Panel Member]
Panel Member

APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development.

Prof. Alpha
Program Chair/Coordinator,
Information Technology

Dean Delta
Dean, School of Computing

Acknowledgment

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Summary of Revisions

The following revisions were made in response to the panel's recommendations recorded in the Capstone Revision Form (FM-AAC-SOC-2004) after the final defense on April 12, 2027.

Panel recommendation	Action taken	Section
Add the per-item evaluation results to the Results, not only the means.	Results now include the full evaluation table per item.	Results
Explain the lowest-rated characteristic in the Discussion.	Discussion now explains the cause of the lowest rating and the fix made.	Discussion
Align the recommendations with the limitations found.	Each recommendation now answers a limitation found during testing.	Recommendations

Checked by: _____ Prof. Charlie, Adviser

Smart Queue Management System for University Clinics

November Student, Oscar Student, Hotel Student, India Student

Holy Angel University
Information Technology

Abstract

This study aimed to develop and evaluate a web-based Smart Queue Management System for the university clinic. Patients cannot see their position in line, staff are interrupted by repeated questions, urgent cases wait behind routine clearances, and the clinic cannot report service volume. The Smart Queue system made the clinic queue visible to patients and staff, reduced waiting time in the simulated session, and was rated excellent in usability (SUS 84.0). Average waiting time fell from 41 to 26 minutes in the simulated session; no patient left before being served.

Keywords: capstone, web application, ISO/IEC 25010, usability

SMART QUEUE MANAGEMENT SYSTEM FOR UNIVERSITY CLINICS

Introduction

University clinics serve a steady stream of students, faculty and staff who arrive for consultations, medical clearances, dental check-ups and the release of laboratory results. At most campuses the arrival of these patients is still managed by a paper logbook at the reception desk and a verbal call of names once a nurse or physician is free.

Digital queue management systems address this by issuing a numbered ticket, displaying the current number, and notifying the patient when their turn approaches. Commercial systems are priced and designed for large institutions; a lightweight web-based system running on the phones students already carry would suit a campus clinic better.

The problem this study addresses is the following: patients cannot see their position in line, staff are interrupted by repeated questions, urgent cases wait behind routine clearances, and the clinic cannot report service volume. This capstone project therefore proposes the Smart Queue system.

Review of Related Literature

This section reviews studies and existing systems related to the Smart Queue system and the methods used to evaluate it.

Queueing Theory and Perceived Waiting Time

For a single-server clinic desk, the expected waiting time grows sharply as utilization approaches one (Example & Writer, 2022). Uncertain waits feel longer than known, finite waits of the same duration (Placeholder, 2021), which supports giving every patient a live view of the queue.

Web-Based and Mobile Queue Management Systems

A registrar queue built as a progressive web application reduced counter congestion by letting students wait outside and return when notified (Author et al., 2023). The proposed system adopts ticket issuance without a kiosk, a public display and staff call controls.

Notification Channels and Response Behavior

Email and push notifications are nearly free to send, while SMS carries per-message costs small institutions struggle to sustain (Writer, 2022).

Software Quality Evaluation

Capstone projects in the Philippines commonly ask IT experts to rate a system against ISO/IEC 25010 characteristics on a Likert scale (Example, 2023).

Synthesis

The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the Smart Queue system fills.

Conceptual Framework

The study follows the Input–Process–Output model shown in Figure 1.

Figure 1

Conceptual Framework of the Study

INPUT	PROCESS	OUTPUT
Current clinic process Staff and patient requirements React, Firebase	Requirement analysis Design and development (Scrum) Testing and evaluation	Smart Queue Management System for University Clinics Evaluation results

Objectives of the Study

The general objective of the study is to develop and evaluate a web-based Smart Queue Management System for the university clinic. Specifically, the study aims to:

1. analyze the current patient reception and queueing process of the clinic and identify its problems;
2. design and develop a system that lets patients join the queue through a QR code or the university portal, view their position and estimated waiting time, and receive an email when their turn approaches;
3. provide clinic staff with a dashboard to call, skip, recall and prioritize tickets, and to generate daily and monthly service reports;
4. evaluate the system in terms of functional suitability, performance efficiency, usability and reliability based on ISO/IEC 25010; and
5. measure the perceived usability of the system using the System Usability Scale.

Scope and Delimitations

The system covers the queue for walk-in services of the university clinic: medical consultation, dental consultation, medical clearance and the release of results. Patients sign in with their university email address; a patient may hold one active ticket at a time.

The study does not cover appointment scheduling, electronic medical records, prescriptions, billing or any clinical data beyond the service requested. Notifications are sent by email only; SMS is outside the scope because of its per-message cost.

Method

This section describes how the Smart Queue system will be analyzed, specified, designed, developed, tested and evaluated.

Requirement Analysis

The researchers observed the current process and interviewed its users. The current process is shown in Appendix B.

Requirements Documentation

The functional requirements are grouped by user in Table 1; the non-functional requirements follow in Table 2.

Table 1

Functional Requirements

ID	User	Requirement
FR-01	Patient	Sign in with a university email address
FR-02	Patient	Join the queue by scanning a QR code or from the portal
FR-03	Patient	View position in line and estimated waiting time
FR-04	Patient	Receive an email when two turns away; cancel a ticket
FR-05	Staff	Call the next ticket; skip, recall and complete tickets
FR-06	Staff	Move a ticket to the front with a recorded reason
FR-07	Administrator	Generate daily and monthly reports

Table 2

Non-Functional Requirements

Characteristic	Requirement
Performance	Queue updates reach every open screen within two seconds
Usability	A first-time patient can join the queue in under one minute
Security	Only university email addresses can sign in; staff actions are logged
Reliability	Tickets survive a page refresh or a lost connection

Design of Software, System, Product and Process

The system uses a client–server architecture: a React client, Firebase Authentication restricted to university accounts, Cloud Functions for every write, Firestore for data and a mail service for notifications. The main entities are PATIENT, SERVICE, TICKET, STAFF, STAFF_ACTION.

Development and Testing

Development follows Scrum with two-week sprints. Testing proceeds from unit tests to integration tests on the Firebase Emulator Suite to acceptance testing with the functionality checklist (Appendix C).

Sample and Setting

Respondents are selected by purposive sampling as shown in Table 3.

Table 3

Respondents of the Study

Respondent group	Number	Instrument
Clinic staff	5	SUS, ISO/IEC 25010
Student patients	30	SUS
Faculty and employee patients	10	SUS
IT experts	5	ISO/IEC 25010

Data Analysis Plan

SUS scores are computed per respondent and interpreted with the adjective scale. ISO/IEC 25010 ratings are summarized by weighted mean (4.21–5.00 Strongly Agree; 3.41–4.20 Agree; 2.61–3.40 Neutral; 1.81–2.60 Disagree; 1.00–1.80 Strongly Disagree).

Ethical Consideration

Participation is voluntary and respondents may withdraw at any time. Responses are anonymous, kept for the duration of the study and destroyed after the final defense, in accordance with the Data Privacy Act of 2012.

Results

Table 4

Functionality Checklist Results

Function	Result
Sign in with a university email	Working
Join a queue by QR code	Working
View position and estimate	Working
Email two turns away	Working
Call next ticket	Working
Skip and recall	Working
Prioritize with a reason	Working
Public display updates	Working
Daily report	Working
Monthly report	Working

Table 5

ISO/IEC 25010 Evaluation

Characteristic	Weighted mean	Interpretation
Functional Suitability	4.61	Strongly Agree
Performance Efficiency	4.38	Strongly Agree
Usability	4.55	Strongly Agree
Reliability	4.40	Strongly Agree

The mean SUS score was 84.0, which is above the benchmark of 68.

Simulated morning. Average waiting time fell from 41 to 26 minutes in the simulated session; no patient left before being served.

Discussion

Clinic staff valued the priority control most; it let them serve urgent cases without arguments at the desk. Students rated the estimated waiting time as the most useful patient feature, although several noted that the estimate jumps when staff take a break, which the group addressed by showing a range instead of a single number.

Conclusion

The Smart Queue system made the clinic queue visible to patients and staff, reduced waiting time in the simulated session, and was rated excellent in usability (SUS 84.0).

Recommendations

1. Add a text-to-speech call on the public display.
2. Integrate with the clinic's appointment calendar once one exists.
3. Pilot the system during the annual physical examination period.

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Appendix A

Cover Letter to Host/Locale

September 14, 2026

The Head Nurse
University Health Services
Holy Angel University

Dear Madam:

We are students of the School of Computing working on our capstone project entitled "Smart Queue Management System for University Clinics." We respectfully ask permission to observe the current process at your office, interview your staff, and conduct a test session of the system. This letter is a placeholder for the dummy manuscript.

Respectfully yours,

November Student
Oscar Student
Hotel Student
India Student

Noted by:
Prof. Charlie, Adviser

Appendix B

Current Process Flowchart

Request is made by message or on paper

Staff checks availability manually

Request is confirmed or declined

Activity takes place

Record is kept on paper, if at all

Appendix C

Functionality Checklist

Function	Working	Not working	Remarks
Sign in with a university email			
Join a queue by QR code			
View position and estimate			
Email two turns away			
Call next ticket			
Skip and recall			
Prioritize with a reason			
Public display updates			
Daily report			
Monthly report			

Appendix D

System Usability Scale Questionnaire

1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, 5 – Strongly Agree

Statement	1	2	3	4	5
1. I think that I would like to use this system frequently.					
2. I found the system unnecessarily complex.					
3. I thought the system was easy to use.					
4. I think that I would need the support of a technical person to be able to use this system.					
5. I found the various functions in this system were well integrated.					
6. I thought there was too much inconsistency in this system.					
7. I would imagine that most people would learn to use this system very quickly.					
8. I found the system very cumbersome to use.					
9. I felt very confident using the system.					
10. I needed to learn a lot of things before I could get going with this system.					

Appendix E

ISO/IEC 25010 Software Quality Evaluation Form

Criteria	5	4	3	2	1
Functional Suitability					
The system meets the functional suitability requirements.					
Performance Efficiency					
The system meets the performance efficiency requirements.					
Usability					
The system meets the usability requirements.					
Reliability					
The system meets the reliability requirements.					